J. Geomag. Geoelectr., **41**, 101–117, 1989

Multi-Component Remanent Magnetization in the Aulis Volcanics, the Lesser Himalaya, Nepal

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(Received November 10, 1987; Revised August 25, 1988)

Detailed paleomagnetic investigation was carried out on 33 trachyte and trachyandesite rock samples collected at 5 sites from the Aulis volcanics in the Tansen area, the Lesser Himalaya, Nepal. Analysis of magnetic behavior by stepwise alternating field (up to 53 mT) and then thermal demagnetization shows complex multi-component remanence composed of at least four components as follows: 1) Component A (D=001°, I=47°)—of recent field origin, 2) Component B (D=339°, I=32°)—interpreted as acquired during early Miocene, 3) Component C (D=330°, I=−49°)—probably acquired during Late Cretaceous, and 4) Component D, poorly defined, with southerly declination and downward inclination of about 55°—probably acquired prior the acquisition of component C. All these components show in situ grouping suggesting their secondary nature.

Thermomagnetic analysis showed double-Curie point curves indicating two major magnetic phases with Curie temperatures around 565°C and 360°C. These phases are interpreted as of Ti-poor magnetite and titanomaghemite, respectively, which were formed due to low-temperature oxidation of original titanomagnetite. Analysis of unblocking temperature spectra suggests that the 360°C phase contributes to the component B magnetization.

1. Introduction

This study is a continuation of paleomagnetic research into the Nepalese Lesser Himalayan rocks in an effort to augment the sparse paleomagnetic data base available (YOSHIDA and SAKAI, 1984; GAUTAM, 1987). The purpose of this paper is to discuss the magnetic properties of the only known volcanic rocks in the region, with emphasis on their behavior during various kinds of demagnetization techniques. The multi-component nature of these rocks was studied in detail.

2. Geologic Setting and Sampling

The study area belongs to the southern part of the Lesser Himalayan zone just north of the Main Boundary Thrust (MBT) in the vicinity of the town of Tansen in Central West Nepal. Many scientists have contributed to the geological knowledge

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of this area. However, only a brief geological outline based on the recent work by SAKAI (1983), relevant to the present study, will be given here.

Two major lithostratigraphic divisions are recognized in the area: the Kali Gandaki Group and the disconformably overlying Tansen Group. The Kali Gandaki Group represents a sequence of non- to weakly-metamorphosed terrigenous and carbonate rocks, and is inferred to be of late Precambrian to early Paleozoic age. The Tansen Group consists mainly of clastic sediments, and five lithostratigraphic units are recognized there (Table 1).

The Tansen Group forms a synclinal structure known as the Tansen Synclinorium. The central part of the synclinorium is covered by the Palpa Klippe, which is believed to have been derived from the Kali Gandaki Group to the north due to its overthrusting upon the synclinorium.

Volcanic rocks occur in the lower part of the Taltung Formation. Altogether three flows are believed to exist. The flows attain up to 200 m of thickness and are distributed discontinuously in a narrow strip in the southern part of the Tansen Synclinorium (Fig. 1). The lava flows occur parallel to subparallel with the statification planes of the adjacent sedimentary units, and their clasts are also found in conglomerate layers which occur at different levels within the Taltung Formation. This suggests that the volcanic activity occurred during the deposition of the Taltung Formation. The Taltung Formation yields a floral assemblage of *Ptilophylum* sp., *Pterophylum* sp., *Cladophlebis*, sp., *Elatocladus* sp., and Pteridospermae comparable to the Rajmahal flora in Peninsular India. Hence the volcanic rocks, together with the Taltung Formation, are correlated with the Rajmahal traps and the intertrappean sediments of the Indian Peninsula.

Table 1. Lithostratigraphic description of the Tansen Group (after SAKAI, 1983).

Lithostratigraphic unit	Main lithology	Thickness	Age
Dumri Formation	quartzose sandstone, red and green shale	725 m	Oligocene to early Miocene
Disconformity			
Bhainskati Formation	black shale with nummulitic limestone, hematite beds	160 m	Eocene
Amile Formation	white quartzite, ferruginous quartzite, pink limestone	280 m	Late Cretaceous to Paleocene
Disconformity			
Taltung Formation	olive green sandstone, siltstone, conglomerate and volcanics	250 to 300 m	Late Jurassic to early Cretaceous
Disconformity			
Sisne Formation	diamictite, slate and bioturbated mudstone	20 to 1000 m	Late carboniferous to Permian

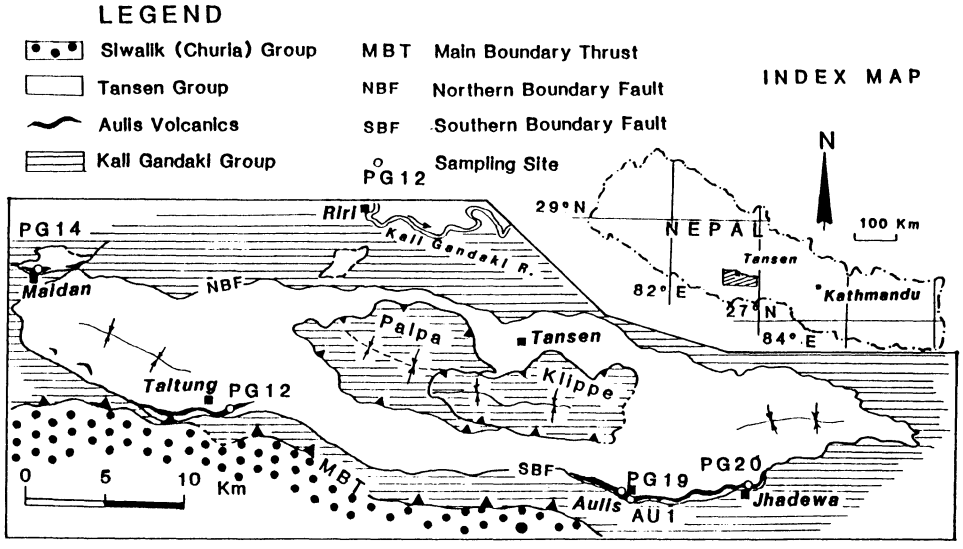


Fig. 1. Geological sketch map of the Tansen area (adapted from SAKAI, 1983) showing sampling sites of the Aulis Volcanics.

Several localities of occurrence of the Aulis volcanics were visited. However, fresh in situ outcrops could be found at only a few places. Samples were collected from 5 sites. Block samples were oriented by using a hand magnetic compass. The sampling sites are shown in the geological sketch map (Fig. 1) and brief sampling information is presented in Table 3.

The samples are light gray to dark-green and sometimes yellowish gray in appearance but have similar lithologic character. They are porphyritic due to phenocrysts of plagioclase and potassium feldspars. Microscopic examination reveals distinct trachytic texture. According to their chemical composition, they belong mostly to the trachyte to trachyandesite compositional range. Due to this reason the term "Aulis Volcanics" seems to be more appropriate than the name "Aulis Basalt" originally proposed by SAKAI (1983). Results of detailed chemical analysis of these rocks, together with their petrological characteristics, are to be published elsewhere (Arita *et al.*, in preparation). Typical analytical data obtained from electron probe microanalysis (EPMA) for one sample is given in Table 2 (S. Okamura, personal communication).

Table 2. Electron probe microanalysis data for sample AU1-4.

Oxide	SiO_2	TiO_2	Al_2O_3	FeO*	MnO	MgO	CaO	Na_2O	K_2O	Total
wt%	62.38	1.03	18.17	4.32	0.11	1.40	1.16	4.85	6.88	100.30

*Total iron as FeO.

Table 3. Sampling details with NRM intensity data. Rock type: Trachyte, Trachyandesite.

Site	Details					NRM intensity, mA/m**		
	Lat. (N) /Long. (E)	Description	Outcrop thickness	Trend /Dip	$n/N/N_d$*	Min.	Max.	Average
AU1	27°45.9′ /83°38.0	Left bank of R., south of Aulis, lower flow?	5 m	295° /55°N	5/17/17	1.39	41.0	16.10
PG12	27°49.3′ /83°21.4′	Right bank of Taltung R. at its lower reaches.	>30 m	255° /35°N	7/16/12	45.40	468.0	256.00
PG14	27°54.2′ /83°13.8′	NW slope of a hill at NW corner of Maidan, alt. 1050 m.	7 m	220° /not det.	7/ 7/ 7	0.23	3.7	1.29
PG19	27°46.7′ /83°36.8′	Near water source, S slope of Aulis, alt. 1535 m, upper flow?	15 m	310° /45°N	7/20/ 7	1.33	65.4	11.90
PG20	27°46.9′ /83°41.3′	NE corner of Jhadewa valley, base of NE-SW flowing R., right bank.	10 m	245° /35°N	7/16/14	4.03	522.0	177.00

*n: number of samples, N: number of specimens, N_d: number of specimens subjected to demagnetization studies.

**NRM intensity data are given at N specimen levels.

3. Laboratory Techniques

In the laboratory, specimens in the form of cylinders 2.54 cm in diameter and 2.4 cm in length were cut for measurements of intensity and direction of remanent magnetization. Remanent magnetizations were measured by means of Schonstedt SSM-1A spinner magnetometer. Stepwise alternating field (AF) and thermal (TH) demagnetizations were carried out by using a 400-Hz AF demagnetizer and heating furnace, respectively, both being placed inside individual sets of Helmholtz coils to compensate for the ambient geomagnetic field in the laboratory.

Thermomagnetic analysis was carried out on whole-rock (chip) samples by means of a vertical balance equipped with an automatic recording system. The samples were cycled from room temperature up to 630° C and back in air-free space (vacuum close to 0.26 Pa.) with a heating-cooling rate of 10° C/min. All the measurements were carried out at the paleomagnetic laboratory of Hokkaido University.

4. Results of Measurements

4.1 Natural remanent magnetization (NRM) data

NRM intensities for individual specimens ranged from 0.23 mA/m to 522.0 mA/m (Table 3). Every site is characterized by its specific intensity range. A pronounced between-sample variation within a site (by as much as 130 times) can be observed. Directions of NRM were typically highly scattered, indicating the unstable or multivectorial nature of the remanence.

4.2 Pilot demagnetization studies

The first samples subjected to pilot demagnetization studies are those from site AU1. AF demagnetization of one specimen from each of the five samples with 5–10 mT steps, up to a maximum of 50 mT and sometimes to 53 mT, revealed that the collection consists of two subgroups which differ in demagnetization behavior. Specimens from the first subgroup designated as AU1(A) showed that after 15 mT treatment the resultant vectors continue to decay, defining stable end-points with S-SW declinations and moderately steep downward inclinations. On the contrary, the specimens from another subgroup, designated as AU1(B), showed that one or two northerly directed components with moderately downward inclinations became eliminated in general below 40 mT, after which the resultant vectors follow different trajectories indicating that no stable directions are achieved.

Thermal demagnetization was carried out on specimens already subjected to AF demagnetization as well as on virgin specimens, with heating steps of mostly 75–100° C. The already AF-treated specimens from AU1(A) showed no directional changes in the resultant vector up to 550° C, in contrast to those from AU1(B) which exhibited gradual but continuous changes up to 550° C, after which the specimens exhibited random behavior due to the build-up of viscous magnetization, probably

related to chemical changes. A typical example of demagnetization response of a specimen AU1-4A representing the latter group is shown in Fig. 2. It is clear that the specimen possesses at least three components, of which the softer two components (A and C, definition of the components is described later) have clearly non-overlapping demagnetization spectra (one within AF range, the other within TH range). The convergence to the end-point (third component) is not so clear, but the resultant vector direction in final steps was found to be close to the single component present in subgroup AU1(A) specimens. Thermal demagnetization of virgin specimens AU1(B) showed less straightforward vector trajectories, apparently due to overlapping of the unblocking temperature spectra of the first two components. Figure 3 demonstrates the results of comparative study of the behavior of specimens from site PG20 during AF-followed-TH demagnetization and TH only demagnetization. These specimens had initial intensities of the same order or magnitude. In the case of AF-followed-TH demagnetization (PG20-5B), it is evident that a viscous component became eliminated during the initial stages of AF demagnetization, and upon further demagnetization the other two components (A and C) can be clearly identified. In the case of TH-only demagnetization (PG20-7B), however, the identification of these components is not so easy, although their existence cannot be denied. Hence the AF-followed-TH demagnetization pro-

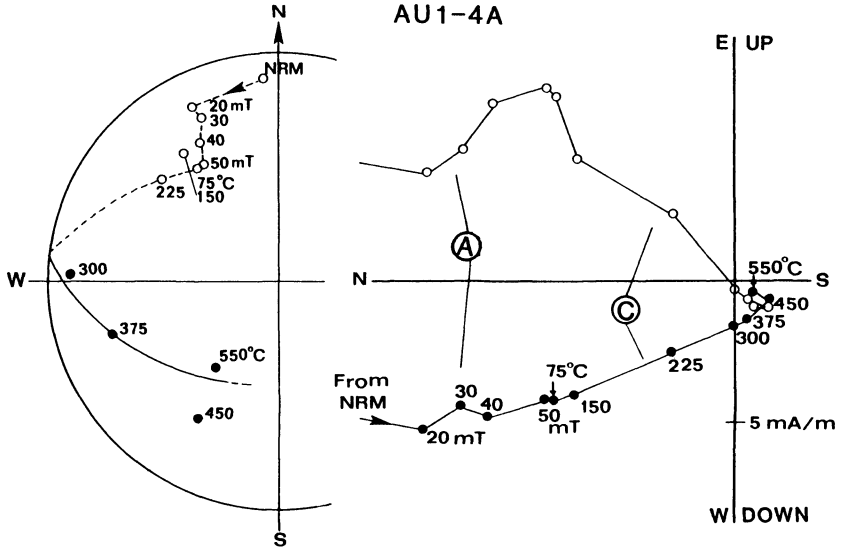


Fig. 2. AF-followed-TH demagnetization results for a representative specimen AU1-4A. Left: vector migration plot. Solid (open) circles denote downward (upward) inclinations. Right: orthogonal projection diagram (ZIJDERVELD, 1967) showing projections of the end-points of the resultant vector during successive demagnetization. Solid and open circles are the projections on the horizontal and N-S vertical planes, respectively. The alphabets, e.g., A, C, enclosed with circles denote the components reflected in the linear segments opposite which they are placed.

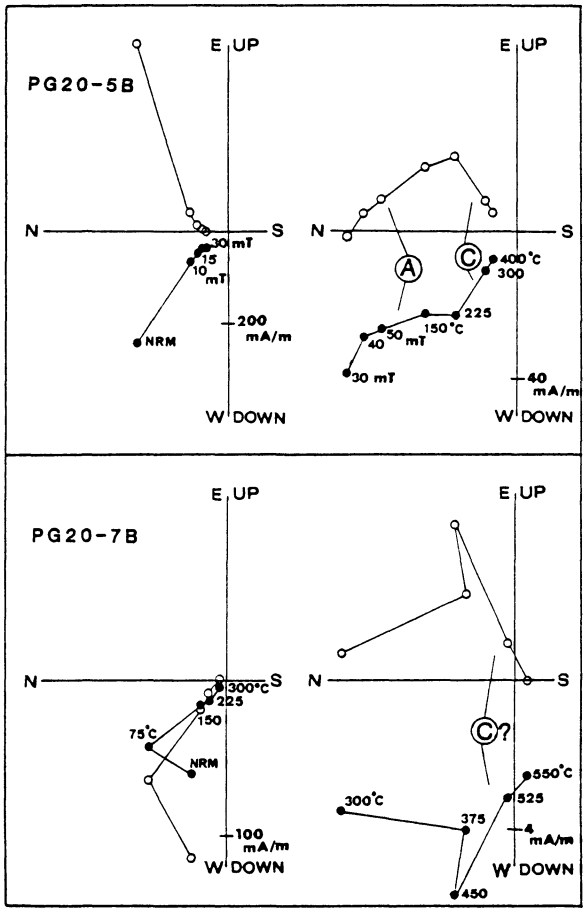


Fig. 3. Orthogonal projection diagrams for comparison of the magnetic behavior of specimen PG20-5B during AF-followed-TH demagnetization (upper diagram) with that of specimen PG20-7B during TH demagnetization (lower diagram). Note that each of the diagrams is divided into two parts (left and right) to adjust the intensity scale. Other conventions as in Fig. 2.

cedure was preferred for routine demagnetization. Low-temperature demagnetization (OZIMA *et al.*, 1963) was tested on several specimens but was found ineffective.

4.3 Routine demagnetization results

Routine AF-followed-TH demagnetization of the specimens showed very complex magnetic behavior which can be summarized as follows:

4.3.1. A multi-component nature of magnetic remanence

It is composed of at least four distinct magnetic directions which are reflected as nearly linear segments in the orthogonal projection diagrams. They are: a northerly

striking component with a downward inclination of about 45°; a NNW-striking component with about a 30° downward inclination; another component also striking NNW but with a steep upward inclination of about 45°; and finally a southerly striking component with a steep downward inclination of about 55°. These directions are hereby designated as components A, B, C and D respectively. The number of components present or determinable in individual specimens varied from only one to all of the four mentioned.

4.3.2. A general nonuniqueness in the component hardness spectra (coercivity or unblocking temperature ranges)

The normal demagnetization sequence observed in most of the specimens was A-B-C-D. Components A and B are usually completely eliminated within AF range. Component C has thermally distributed unblocking spectra typically in the range of 225–500° C, as opposed to the discrete spectra for component D very close to 550° C or above. However, various hardness sequences such as A-C-A-C, A-D-C, or A-D were frequently observed (Fig. 4). A notable feature is that Component B always has blocking temperature ranges below 350–360° C.

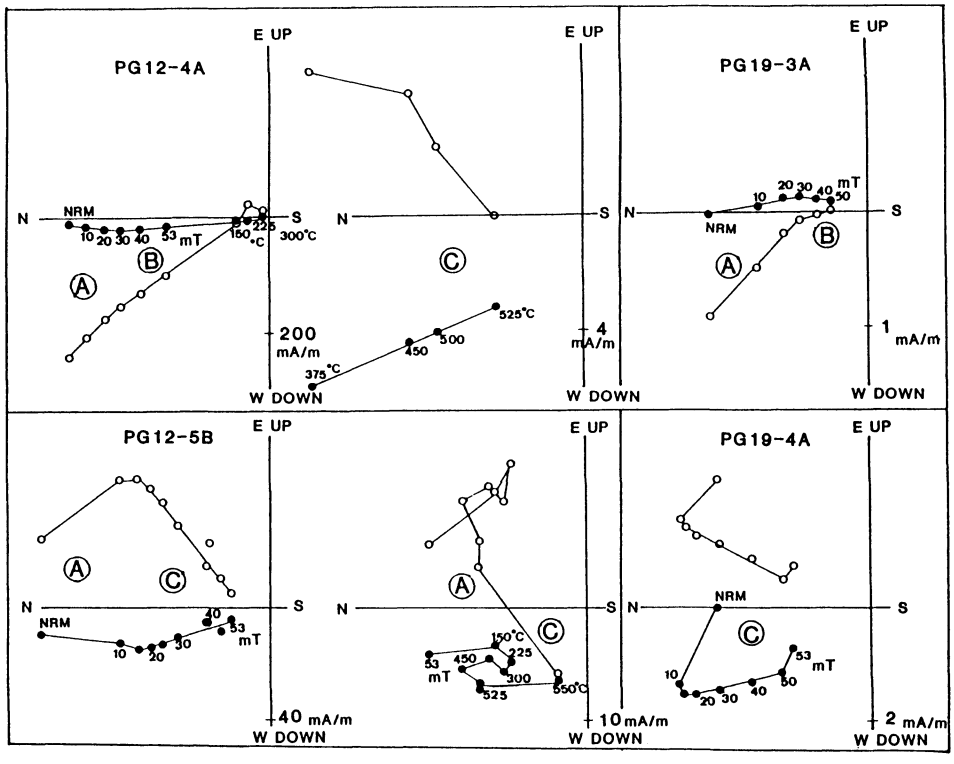


Fig. 4. Orthogonal projection diagrams for specimens PG12-4A, PG12-5B, PG19-3A and PG19-4A representative for the rocks studied.

4.3.3. *Nonunique intensity response for various specimens during demagnetization*

The highly variable NRM intensities at specimen level, the varying number of components present in each of them, and the nonuniqueness in the component hardness spectra are among the interrelated factors which contributed to the nonidentical intensity responses. This is illustrated in Fig. 5, in which, for the purpose of clarity, the data-points corresponding to the specimens from the same site are labeled with similar symbols. No generalizations could be made regarding to the percentage contribution of individual components to the NRM intensity.

The complex nature of the remanence as outlined above required a nonunique approach to the resolution of the individual components. The criteria of maximum grouping was the only solution for specimens from site PG14 (Fig. 6) and subgroup AU1(B). These groups were dominated by components C and D, respectively. For the rest of the data, a two fold approach for line-analysis was applied: first, the data were subjected to linearity spectrum analysis (LSA) to separate collinear data-point segments (SCHMIDT, 1982), and then subsequent least-square line analysis of the segments (KIRSCHVINK, 1980) was conducted. In practice, the linearity coefficients for meaningful segments were well above 0.99, and they were accepted if the best-fit line had an error-angle (MAD) less than 10°. Remagnetization circle plane analysis was carried out also using the least-square technique for datasets which showed the planar character (in vector migration plots), and the subsequent poles were analyzed by the metho of BAILEY and HALLS (1984) to determine the best estimates of component D.

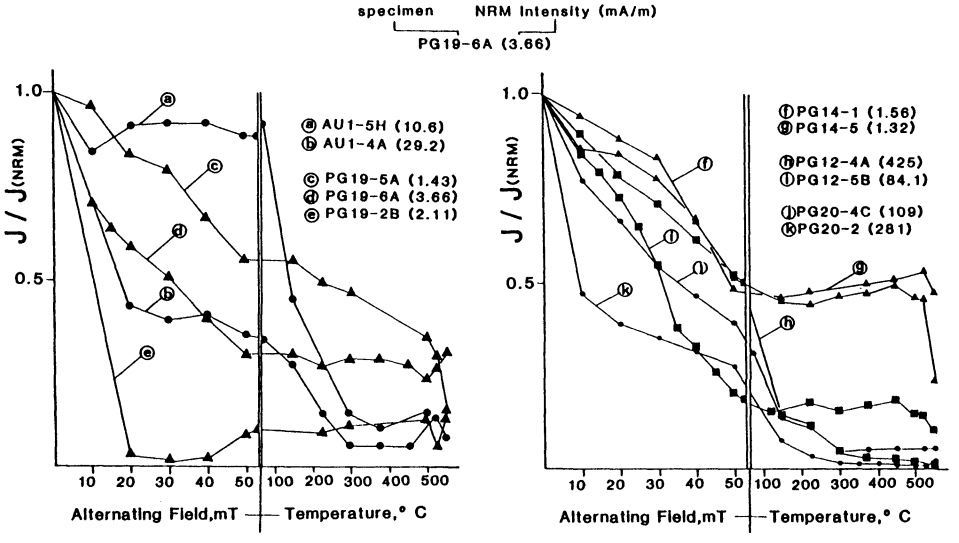


Fig. 5. Normalized intensity response curves to show the variation in the intensity changes during demagnetization.

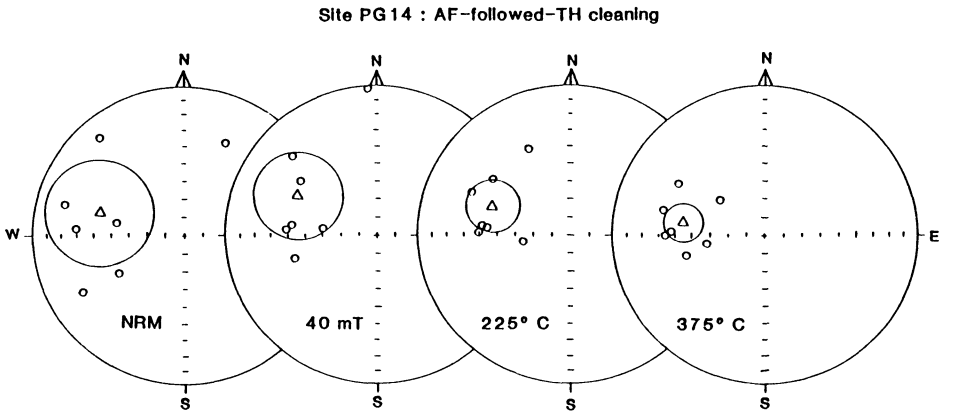


Fig. 6. Progression of grouping of the directions (circles) after successive cleaning in specimens (samples) from site PG14. Triangles are mean directions and are accompanied by 95% confidence circles.

The results of component-analysis are presented in Table 4, and the respective component plots are shown in Fig. 7. The directions plotted in these diagrams are in geographic coordinates because tilt-correction did not yield better grouping. For component D, the peak-field resultant vectors (squares) at specimen level for subsite AU1(A) clearly indicate a great circle trend which passes through the region defined by the two estimates (triangles) derived from pole data from site PG12 and subsite AU1(B).

4.4 Magneto-mineralogy

Figure 8 shows representative thermomagnetic curves for the chip samples. A double-Curie point behavior during heating is clear. In the majority of curves, the lower and upper Curie temperatures calculated graphically correspond to about 350° C and 565° C, respectively (left diagram, Fig. 8). The cooling curve has a single high Curie-point. A few specimens show the kink-feature around 185° C associated with the lower Curie-point, in which the higher Curie-point falls close to 590° C (right diagram).

Microscopic observations of polished specimens and thin sections showed that relatively large opaque grains (20 to 300 microns) are very rare. They are in a highly altered state characterized by the presence of a polyphase assemblage with regions of varying reflectivity. Some of the grains are characterized by a well-developed radial pattern of microfractures. Such grains possess irregular boundaries and red stains can be observed along the fractures or in the matrix. These observations suggest the circulation of (hydrothermal?) solutions and the migration of iron cations from the opaque grains to the matrix.

EPMA data on the large grains yield compositions consistent with ilmenite-hematite series minerals (Ferrian ilmenite or titano-hematite) occurring mostly in

Table 4. Paleomagnetic results.

Site	$N(n)$[1]	Method[2]	AF or Temperature spectra[3]	Mean Direction[4] D (°)	I (°)	k	a_{95} (°)	VGP[5] Lat. (°N)	Long. (°E)	dp (°)	dm (°)	Paleolatitude[6] (°)
Component A: Recent												
AU1(B)	3(2)	LSA/LSQ	0–40 mT	004	51	20	17.9					
PG12	3(3)	LSA/LSQ	<225°C	003	45	395	4.0					
PG19	3(2)	LSA/LSQ	0–40 mT	004	46	26	15.7					
PG20	5(4)	LSA/LSQ	0–50 mT	351	48	41	9.7					
Mean (4 sites)				001	47	263	4.3					28±2
Component B: early Miocene												
AU1(B)	3(2)	LSA/LSQ	20–40 mT	323	38	1058	2.4	55	350	1	2	21±1
PG12	10(6)	LSA/LSQ	0–50 mT	353	32	33	7.6	77	297	4	8	17±3
PG19	4(3)	LSA/LSQ	10–30 mT	333	28	19	17.1	61	332	10	18	15±8
PG20	9(7)	LSA/LSQ	<300°C	344	28	34	8.0	71	310	4	8	14±4
Mean (4 sites)				339	32	44	10.5	68	329	6	11	17±5
Component C: Late Cretaceous?												
AU1(B)	5(3)	LSA/LSQ	75–225°C	324	−52	44	9.4	20	295	8	12	−33±4
PG12	10(6)	LSA/LSQ	225–550°C	340	−48	21	10.1	29	283	8	13	−30±5
PG14	7(7)	Peak Temp.	375°C	280	−45	20	11.7	—	—	—	—	−27±5
PG19	6(6)	Peak Temp.	300°C	326	−39	32	10.1	30	300	7	12	−22±5
PG20	6(5)	LSA/LSQ	225–500°C	330	−54	28	10.7	21	290	10	15	−35±5
Mean (4 sites excluding PG14)				330	−49	98	7.0	25	292	6	9	−30±3
Component D (Reverse Polarity)												
AU1(A)	5(2)	Peak AF	20 mT	—directions lie along great circle—								
AU1(B)	3(3)	RCM	225–500°C	178	53	150	ASD=4.7					
PG12	4(4)	RCM	225–500°C	164	54	290	ASD=3.4					

1) N and n denote the number of specimens and corresponding samples, respectively. 2) LSA: linearity spectrum analysis, LSQ: least-square line method. RCM: remagnetization circle method. AF: alternating field values. 3) Only dominant spectral ranges are given. 4) Mean directions are calculated giving unit weight to specimens I: inclination, k: precision parameter, a_{95}: 95% confidence radius. ASD: angular standard deviation (for pole data). The directions presented are in situ. 5) VGP: virtual geomagnetic (north) pole position calculated from the mean direction, dp and dm are the 95% confidence ovals of the pole position. 6) Paleolatitudes for sites.

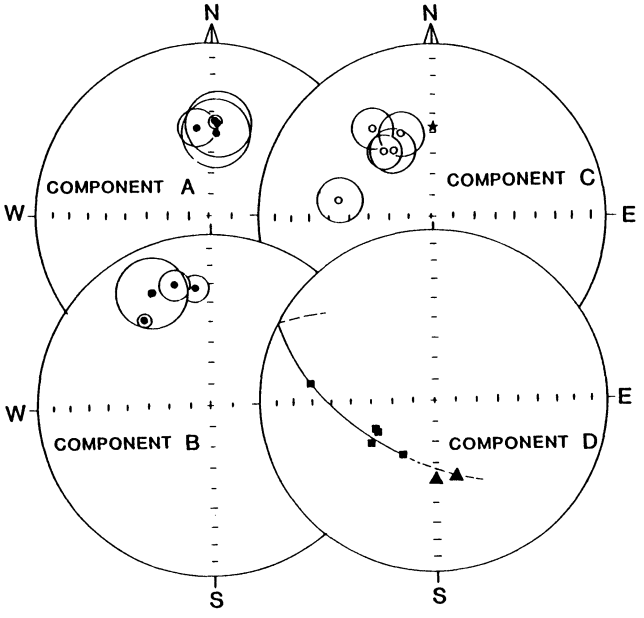


Fig. 7. Results of component-analysis. Solid and open symbols denote downward and upward inclinations, respectively. Small circles are mean directions at site levels and are accompanied by their 95% confidence circles. For component D, triangles are the best estimates from pole data for site PG12 and subsite AU1(B), whereas the squares are the 20 mT cleaning vectors at specimen levels for subsite AU1(A). Star is the dipolar field for Tansen. For data, refer to Table 4.

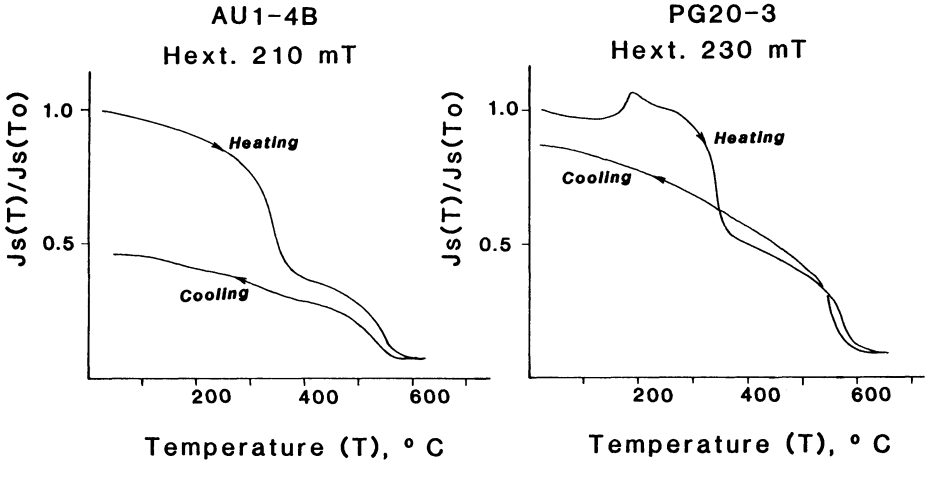


Fig. 8. Representative thermomagnetic curves (specimens AU1-4B and PG20-3).

association with rutile or sphene. A polycrystalline aggregate of rutile and ilmenite or woven alteration of ilmenite-hematite minerals can be observed. A golden yellow phase of sphene and another phase with a distinct white reflection (probably leucoxene) was also observed in some of the sections studied. Unfortunately, the majority of opaques, which are probably the most important from the magnetic point of view, are small (less than 5 microns) and highly altered such that optical observations and EPMA is difficult.

The observed thermomagnetic behavior and findings of microscopic observations (combined with EPMA data) on these specimens can probably be best explained as a consequence of low-temperature oxidation, because these features seem to be very similar to the findings reported by ADE-HALL *et al.* (1971) and BUCKER *et al.* (1986). As these authors suggest, the lower Curie-temperature (360° C) phase with or without the "kink" feature is interpreted to be titanomaghemite. The high Curie-temperature (565° C) phase is most likely to represent a Ti-poor magnetite. The mineral phase with Curie-temperature around 590° C, which was found in only a few cases, may represent a cation-deficient magnetite formed as a result of continued low-temperature oxidation of initial titanomagnetite as suggested by NGUYEN and PECHERSKY (1986).

Several specimens were given artificial TRM by heating them above 600° C or 700° C and cooling in the geomagnetic field conditions of the laboratory. The acquired directions were perfectly parallel to the ambient field, and the intensities of TRMs in these specimens were one to three orders higher than the initial NRM intensities. In contrast to the pronounced between-specimen variation of NRM intensities (Table 3), the TRM intensities differed by only one order of magnitude. The high-temperature oxidation apparently resulted in homogenization of the magnetic phase such that it becomes capable of carrying very stable TRM. Figure 9 shows the results of thermal demagnetization of a specimen from site PG20 which had been given an artificial TRM.

5. Interpretation and Discussion

Component A is very close to the present dipolar field (I=47°, for Tansen) and most probably represents viscous remanence.

The interpretation of other components, at present, does not seem straightforward owing to the lack of time-constrained paleomagnetic data base from the region. The preliminary paleomagnetic direction, derived from the Aulis Volcanics by YOSHIDA and SAKAI (1984) was based on 20 mT cleaning from one site and seems to be the result of an unknown number of components. For tentative interpretation of the components B and C, a paleolatitude-time plot for the locality of Tansen was constructed by using the apparent polar wander path (APWP) for India (Fig. 10). As mentioned by KLOOTWIJK *et al.* (1985), there exist three independently simulated APWP trajectories for India which are in good mutual agreement. For the present purpose, the simulated APWP based on India–Africa relative movement with Africa fixed to the hotspot frame (KLOOTWIJK *et al.*, 1985,

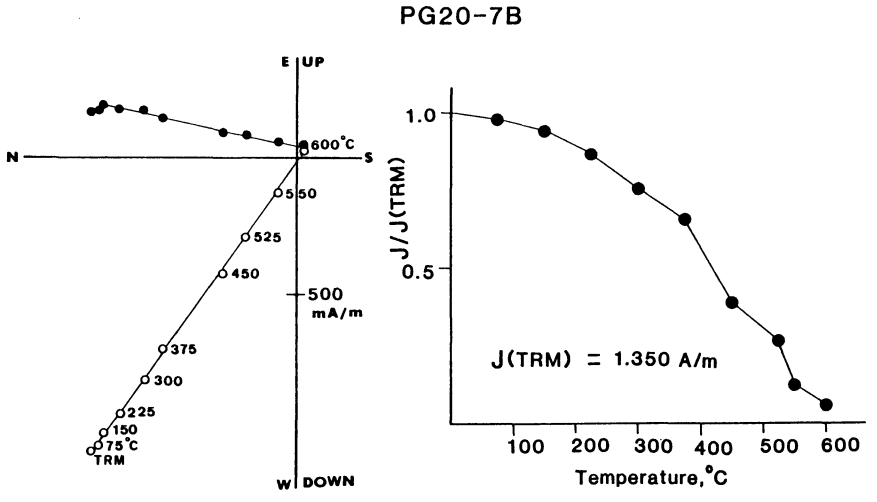


Fig. 9. Orthogonal projection diagram (left) and normalized intensity response curve (right) of specimen PG20-7B during TH demagnetization, after the specimen had been given an artificial TRM in the earth's magnetic field. For comparison with the behavior during TH demagnetization of NRM of the same specimen refer to Fig. 3 (lower diagram).

Table 2C) was used owing to its detail covering the period from 80 Myr to 10 Myr. The underlying assumption is that a hypothetical point at Tansen formed part of the Indian plate and moved northward together with it. The paleolatitude-time plot in Fig. 10 includes an additional point for 102.5 Myr which was calculated by using the pole estimate for the Rajmahal traps of estimated age range 100–105 Myr (KLOOTWIJK, 1971).

The paleolatitude estimates from the mean site inclinations forming components B and C are plotted in such a way that they form crosses centered in the calculated curve. The vertical and horizontal bars of the crosses are errors in paleolatitude estimation (calculated from a_{95} values) and the corresponding age ranges at site level respectively. Such a comparison yields a most probable age of acquisition of Component B to be around 20–25 Myr and that of component C around 75–80 Myr. The first estimation is not so far from the Potassium-Argon (K/Ar) whole rock dating age of 17.4±0.9 Myr obtained from a single specimen from the Aulis Volcanics (K. Arita, personal communication). An early Miocene metamorphic and/or thrusting event may have caused secondary heating of the rocks, giving way to chemical changes of titanomagnetite to form a new phase of titanomaghemite capable of acquisition of thermo-chemical remanence. The second estimation suggests that the rocks were tilted before 75–80 Myr as opposed to the assumption of the post-early Miocene folding of the Tansen Group (SAKAI, 1983).

Component D is nearly antipodal to component C. So, it has reverse polarity. Contrary to component C, this component has higher unblocking temperatures. Its

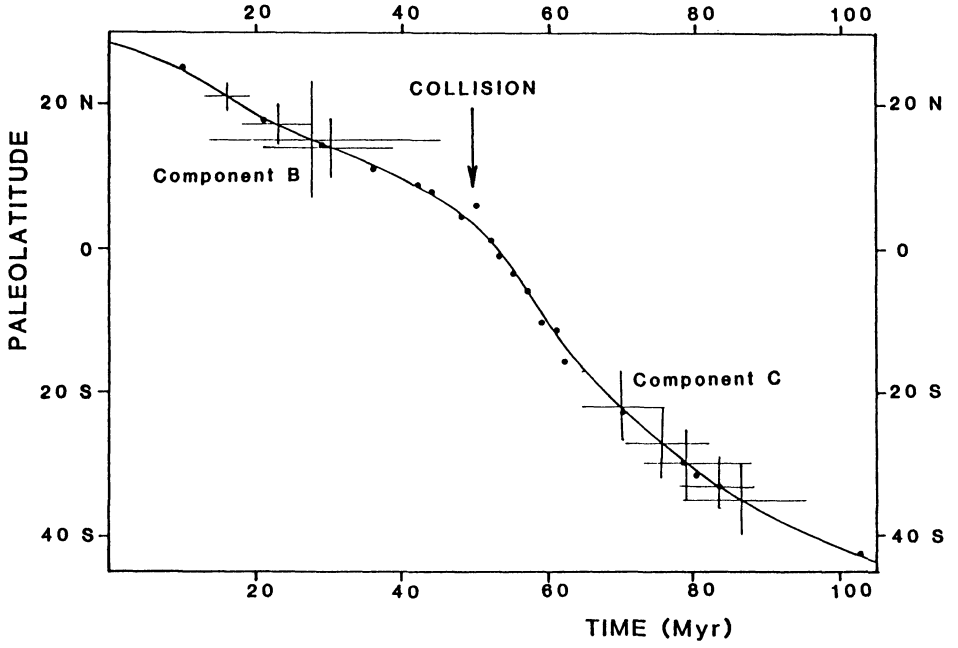


Fig. 10. Paleolatitude-time plot for the town of Tansen. Dots are the expected paleolatitudes as a function of time, and continuous curve is the approximated curve. The crosses are the derived paleolatitudes from the practical data. Detailed explanation is given in the text.

interpretation seems more difficult, but the possibility that this magnetization was acquired prior to the acquisition of component C during a reverse polarity interval can be pointed out.

6. Conclusion

The main finding of this study is that the Aulis volcanics in the Lesser Himalaya possess a highly complex multi-component remanent magnetization. The rocks have experienced at least three magnetization/remagnetization episodes besides the recent magnetization. These magnetic components seem to rest on distinct mineralogical phases produced chemically from the original titanomagnetite contained in the rocks during initial cooling. These components can be resolved by an integrated cleaning approach such as AF-followed-TH demagnetization.

The Aulis Volcanics, though of limited distribution, seem to be representative rocks for understanding the magnetomineralogical consequences of low-temperature oxidation. These rocks also show that the process of alteration and its magnetic consequence vary in accordance with the micro-environmental conditions, which in turn are time-dependent (STORETVEDT, 1973).

An unambiguous interpretation of the various components seems to be rather difficult at present because of the lack of reference data. The lack of knowledge of timing of various events which are likely to contribute to thermo-chemical effects, which in turn may have paleomagnetic significance, seems to be the major obstacle. The uncertainties encountered in this study and the solution of the problems discussed may require not only a large paleomagnetic data base from the Lesser Himalaya itself, but also an integrated and multidisciplinary approach. Further magnetomineralogical analysis of the Aulis Volcanics and paleomagnetic study of the sedimentary rocks from the Tansen Group is in progress and is believed to contribute to the better interpretation of the components involved.

The author gratefully acknowledges the continuous guidance and encouragement of Prof. M. Kato and Associate Professor Y. Fujiwara, of Hokkaido University, throughout this work. He is much indebted to Dr. K. Arita and Dr. M. Yoshida of Hokkaido University, Dr. H. Sakai, of the Earthquake Research Institute, Tokyo, and Dr. S. Okamura, of Hokkaido University of Education, Sapporo College, for valuable discussions. He would like to thank Prof. H. C. Halls of Toronto University for kindly sending the program for pole analysis and Mr. K. Fukuma, of Kyoto University, who assisted by lending the program for linearity and least-squares analyses of demagnetization data.

Sincere thanks are extended to Dr. M. P. Sharma, Mr. M. R. Dhital and other colleagues from Tribhuvan University, Kathmandu, for their advice and encouragement in various ways. The author acknowledges the receipt of the Monbusho (Ministry of Education, Japan) Research Student Scholarship, which made this work possible.

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